

Package ‘tauPanel’

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Type Package

Title Fast Tau Estimation for Fixed and Random Effects Panel Data Models

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Description Implements robust fast Tau estimators for balanced linear panel data models under fixed- and random-effects specifications. The fixed-effects estimator is based by default on first differencing, which eliminates time-invariant individual effects while limiting the propagation of contaminated observations through the panel transformation. The random-effects estimator uses a two-stage procedure combining a difference-based fast Tau pilot fit, robust scale-based variance-component estimation, residual-based observation weights, and robust quasi-demeaning. Computation extends fast Tau regression through cluster-diverse randomized starting subsamples and iterative reweighted least-squares refinement. The package provides unified and model-specific fitting interfaces, structural-mean prediction methods, computational controls, and simulation tools for generating panel data under clean, vertical, leverage, and joint contamination settings.

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stats,
utils

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fast_tau_control

*Default Control Parameters for Fast Tau Panel Estimation***Description**

Returns the default computational and transformation settings used by the fast Tau estimators for fixed- and random-effects panel data models. The defaults correspond to the numerical configuration adopted for the proposed estimators.

Usage

```
fast_tau_control()
```

Details

The proposed panel estimator is based on regression Tau-estimation, in which the coefficient vector is obtained by minimizing a Tau-scale of the transformed regression residuals. The Tau-scale combines a robust M-scale with a second bounded loss function, thereby allowing robustness and efficiency to be controlled separately.

Because the Tau objective is non-convex and may contain multiple local minima, the estimator is computed using a multi-start fast-Tau strategy. Random starting subsets are partially refined by iteratively reweighted least squares (IRLS), promising candidates are retained in a shortlist, and only this shortlist is subjected to more extensive refinement.

For panel data, the default implementation uses cluster-diverse starting subsamples. Distinct cross-sectional units are selected and at most one transformed observation is drawn from each selected unit. This construction reduces the possibility that a starting subsample is dominated by several dependent observations from the same individual. When `m_cluster = NULL`, the number of selected units is set equal to the number of regression coefficients, producing a cluster-diverse elemental start.

The default computational settings correspond to 2000 random starts, two preliminary IRLS updates per candidate, and a shortlist containing the 20 most promising candidates. In the notation of the fast-Tau algorithm, these quantities correspond respectively to $S = 2000$, $\kappa = 2$, and $\ell = 20$. Thus, the component `N` returned by this function denotes the number of random starts and should not be confused with the conventional use of N for the number of cross-sectional units in a panel.

For fixed-effects estimation, the default transformation is first differencing. If

$$y_{it} = x_{it}^T \beta + \alpha_i + \varepsilon_{it},$$

then first differencing gives

$$\Delta y_{it} = \Delta x_{it}^T \beta + \Delta \varepsilon_{it},$$

and therefore eliminates the time-invariant individual effect α_i . In addition to this standard identification property, first differencing has a robustness advantage in the proposed procedure: contamination of a single unit-time observation affects at most two adjacent transformed observations, whereas ordinary within demeaning can propagate its effect through the entire transformed block of that unit.

The argument `approximate` controls how the residual M-scale is evaluated during candidate comparison. With the default `approximate = 0`, the algorithm uses the full M-scale calculation implemented by the fitting routine. When a nonzero value is supplied, a shorter scale approximation is used, with `rr` determining the number of scale-update iterations.

Value

A named list containing the following components:

N	Number of randomized starting subsamples used by the fast-Tau algorithm. The default is 2000. This corresponds to S in the algorithmic notation of the accompanying methodology, and not to the number of cross-sectional units.
kk	Number of preliminary IRLS updates applied to each random or diagnostic starting candidate before candidate screening and shortlisting. The default is 2, corresponding to $\kappa = 2$.
tt	Number of best candidate solutions retained for subsequent refinement. The default is 20, corresponding to a shortlist size $\ell = 20$.
rr	Number of fixed-point scale-update iterations used when an approximate M-scale calculation is requested. The default is 2. This component is relevant when approximate is nonzero.
approximate	Indicator controlling the calculation of the residual M-scale during the candidate-evaluation stage. The default 0 uses the full scale calculation implemented by the fitting routine. A nonzero value requests the shorter approximation based on rr scale updates.
cluster_start	Logical value indicating whether cluster-diverse starting subsamples are used. The default is TRUE. Under cluster-diverse sampling, each starting subsample contains at most one transformed observation from any selected cross-sectional unit.
m_cluster	Number of distinct cross-sectional units used to construct each cluster-diverse starting subsample. The default is NULL, in which case the fitting routine uses the number of regression coefficients, producing an elemental cluster-diverse starting subset. Values larger than the number of coefficients produce overdetermined cluster-diverse starts, subject to the number of available units.
seed	Optional random-number seed used for the randomized starting subsamples. The default is NULL, so no fixed local seed is imposed.
fe_transform	Character string specifying the default transformation for the fixed-effects estimator. The default is "first_difference".

Examples

```
## Obtain the default control settings
control <- fast_tau_control()
control

## Inspect individual settings
control$N
control$kk
control$tt
control$fe_transform

## Control values are returned as an ordinary list and may be modified
## before being supplied to the corresponding fitting function.
control$N <- 1000
control$tt <- 10
```

fast_tau_fe

*Fast Tau Estimation for Fixed-Effects Panel Models***Description**

Fits the proposed robust fast Tau estimator for a balanced fixed-effects panel regression model. By default, the estimator is applied to first-differenced observations, although ordinary within-mean and within-median transformations are also available.

Usage

```
fast_tau_fe(
  Y,
  X,
  transform = c("first_difference", "within_mean", "within_median"),
  N = 2000,
  kk = 2,
  tt = 20,
  rr = 2,
  approximate = 0,
  cluster_start = TRUE,
  m_cluster = NULL,
  seed = NULL
)
```

Arguments

Y	A numeric $n \times T$ matrix containing the panel responses, where n is the number of cross-sectional units and T is the number of time periods. The panel must be balanced.
X	A numeric $n \times T \times p$ array containing the regressors, where p is the number of regression coefficients. The first two dimensions must agree with the dimensions of Y.
transform	Character string specifying the transformation used to eliminate the individual fixed effects. Available choices are "first_difference", "within_mean", and "within_median". The default is "first_difference", which corresponds to the proposed fixed-effects estimator.
N	Number of randomized starting subsamples used by the fast-Tau algorithm. The default is 2000. This quantity corresponds to the number of random starts, denoted by S in the methodological description, and should not be confused with the number of cross-sectional units.
kk	Number of preliminary iteratively reweighted least-squares (IRLS) updates performed for each random or diagnostic starting candidate. The default is 2, corresponding to $\kappa = 2$.
tt	Number of the most promising candidate solutions retained for subsequent refinement. The default is 20, corresponding to the shortlist size $\ell = 20$.
rr	Number of fixed-point scale updates used when an approximate residual-scale calculation is requested. The default is 2. This argument is relevant when approximate is nonzero.

approximate	Indicator controlling calculation of the residual M-scale during candidate evaluation. The default 0 uses the full scale calculation implemented by the fitting routine. A nonzero value uses the shorter approximation based on rr scale updates.
cluster_start	Logical value indicating whether cluster-diverse starting subsamples are used. The default is TRUE. When enabled, no cross-sectional unit contributes more than one transformed observation to a starting subsample.
m_cluster	Number of distinct cross-sectional units used in each cluster-diverse starting subsample. If NULL, the number of units is set equal to the number of regression coefficients, yielding a cluster-diverse elemental start. Larger values produce overdetermined cluster-diverse starting subsets, subject to the number of available units.
seed	Optional integer seed for the randomized starting-subsample generation. The default is NULL. The seed is used locally within the fast-Tau computation.

Details

Consider the balanced fixed-effects panel regression model

$$y_{it} = x_{it}^T \beta + \alpha_i + \varepsilon_{it}, \quad i = 1, \dots, n, \quad t = 1, \dots, T,$$

where β is the vector of regression coefficients, α_i is an unobserved time-invariant individual effect, and ε_{it} is the idiosyncratic error.

The proposed fixed-effects estimator uses first differencing by default. Defining $\Delta y_{it} = y_{it} - y_{i,t-1}$ and $\Delta x_{it} = x_{it} - x_{i,t-1}$ gives

$$\Delta y_{it} = \Delta x_{it}^T \beta + \Delta \varepsilon_{it}, \quad t = 2, \dots, T,$$

so that the individual effect α_i is removed exactly.

First differencing is particularly useful from a robustness perspective. A contaminated observation at an interior time point can affect at most the two adjacent first differences, whereas ordinary within demeaning can transmit the effect of a contaminated observation to every transformed observation belonging to the same cross-sectional unit. Moreover, when the level idiosyncratic errors are identically distributed and independent within a unit, their differences are marginally symmetric about zero even when the original error distribution is asymmetric. Adjacent differences are, however, dependent because they share a common level error.

After transformation, the regression coefficients are estimated by minimizing a Tau-scale of the transformed residuals. For a candidate coefficient vector β , let $e_j(\beta)$ denote the corresponding transformed residuals. The Tau-scale has the general form

$$\tau^2(\beta) = s^2(\beta) \left\{ \frac{1}{mb_2} \sum_{j=1}^m \rho_2 \left(\frac{e_j(\beta)}{s(\beta)} \right) \right\},$$

where $s(\beta)$ is a robust M-scale determined by a bounded loss function ρ_1 , and ρ_2 is a second bounded loss function used to control efficiency. The estimator seeks a coefficient vector having a small Tau-scale.

The implementation uses the optimal bounded loss family and the standard fast-Tau tuning adopted in the accompanying methodology. The underlying calibration combines a high-breakdown M-scale with approximately 95 percent Gaussian efficiency in the classical independent-error regression setting. Because first differencing induces within-unit dependence, this percentage is a tuning

calibration and should not be interpreted as an exact 95 percent efficiency statement for the panel estimator.

The Tau objective is non-convex. To explore its relevant local solutions efficiently, the routine uses multiple starting subsamples, a small number of preliminary IRLS updates, candidate screening and shortlisting, followed by more extensive refinement of the most promising solutions. With `cluster_start = TRUE`, starting subsamples are cluster-diverse: distinct units are sampled and at most one transformed observation is selected from each unit. This respects the grouping of the transformed panel observations and reduces the possibility that a starting candidate is dominated by several dependent observations from one unit.

The optional transformations have the following interpretations:

- "first_difference" uses $y_{it} - y_{i,t-1}$ and $x_{it} - x_{i,t-1}$. This is the default and the transformation used for the proposed fixed-effects estimator.
- "within_mean" subtracts the ordinary unit-specific mean from the response and each regressor.
- "within_median" subtracts the unit-specific median from the response and each regressor, providing a robust alternative to mean centering.

The last two choices are provided as alternative transformation options. The methodology proposed for the fixed-effects estimator uses "first_difference".

Before applying the fast-Tau algorithm, the transformed regressor columns are internally rescaled to improve numerical conditioning. The estimated coefficients are subsequently transformed back to the original regressor scale, so this rescaling does not alter the regression model represented by the returned coefficient estimates.

Value

An object of classes "fast_tau_fe_fit" and "fast_tau_panel_fit". It is a list containing:

beta	Numeric vector containing the estimated fixed-effects regression coefficients on the original regressor scale.
fit	The underlying fast-Tau fit object. This contains the low-level optimization results, including the robust coefficient and scale quantities used by the fitting algorithm.
effect	Character string equal to "fixed", identifying the fitted model as a fixed-effects specification.
transform	Character string identifying the transformation used in fitting.
transformed_data	A list containing the transformed response vector y , transformed regressor matrix x , and cross-sectional unit labels id supplied to the fast-Tau regression engine.

Examples

```
## Generate a small fixed-effects panel
dat <- generate_panel_data(
  N_ = 60,
  T_ = 4,
  K = 4,
  effect_model = "fe",
```

```

    contam_scheme = "none"
  )

  ## Fit the proposed fixed-effects estimator.
  fit <- fast_tau_fe(
    Y = dat$Y,
    X = dat$X,
    N = 100,
    tt = 5
  )

  ## Estimated regression coefficients
  fit$beta

  ## Transformation used
  fit$transform

  ## Structural-mean predictions for the supplied predictors
  pred <- predict(fit, newdata = dat$X)
  dim(pred)

```

fast_tau_panel

*Fast Tau Estimation for Panel Data Models***Description**

Provides a unified interface for fitting the proposed fast Tau estimator under either a fixed-effects or random-effects panel-data specification. Depending on effect, the function dispatches to [fast_tau_fe](#) or [fast_tau_re](#).

Usage

```

fast_tau_panel(
  Y,
  X,
  effect = c("fe", "re"),
  ...
)

```

Arguments

Y	A numeric $n \times T$ matrix containing the panel responses, where n is the number of cross-sectional units and T is the number of time periods. The panel must be balanced.
X	A numeric $n \times T \times p$ array containing the regressors, where p is the number of regression coefficients. The first two dimensions must agree with those of Y.
effect	Character string specifying the panel-data specification. Use "fe" for the fixed-effects fast Tau estimator or "re" for the random-effects fast Tau estimator. The default is "fe".
...	Additional arguments passed to the selected fitting function. For effect = "fe", these arguments are passed to fast_tau_fe ; for effect = "re", they are passed

to `fast_tau_re`. These may include computational controls such as the number of random starts, preliminary IRLS iterations, shortlist size, cluster-diverse starting options, and the random seed.

Details

Consider the balanced linear panel-data model

$$y_{it} = x_{it}^T \beta + \alpha_i + \varepsilon_{it}, \quad i = 1, \dots, n, \quad t = 1, \dots, T,$$

where β is the common regression coefficient vector, α_i is an unobserved individual-specific effect, and ε_{it} is the idiosyncratic error.

The appropriate treatment of α_i depends on the assumed relationship between the individual effect and the regressors. Under a fixed-effects specification, α_i may be correlated with the regressor history. Under a random-effects specification, the individual effect is assumed to be mean independent of the regressors, together with the usual variance-component assumptions.

The function `fast_tau_panel()` does not estimate a separate model of its own. It is a convenience dispatcher:

- If `effect = "fe"`, the call is passed to `fast_tau_fe`. The default proposed fixed-effects estimator uses first differencing,

$$\Delta y_{it} = \Delta x_{it}^T \beta + \Delta \varepsilon_{it},$$

which eliminates the time-invariant individual effect exactly. First differencing also limits propagation of a contaminated unit-time observation to at most two adjacent transformed observations.

- If `effect = "re"`, the call is passed to `fast_tau_re`. The proposed random-effects procedure uses a two-stage construction. A difference-based fast-Tau pilot is first obtained and used to construct robust variance-component estimates and residual-based weights. The original panel is then robustly quasi-demeaned and a second fast-Tau fit produces the final regression coefficient estimate.

Both procedures use the same underlying regression Tau-estimation principle. For transformed residuals $e_j(\beta)$, the estimator minimizes a Tau-scale of the general form

$$\tau^2(\beta) = s^2(\beta) \left\{ \frac{1}{mb_2} \sum_{j=1}^m \rho_2 \left(\frac{e_j(\beta)}{s(\beta)} \right) \right\},$$

where $s(\beta)$ is a robust M-scale based on a bounded loss function ρ_1 , while a second bounded loss function ρ_2 controls the efficiency component of the criterion.

The Tau objective is non-convex, so both the fixed- and random-effects procedures employ a fast multi-start strategy. Cluster-diverse starting subsamples, preliminary iteratively reweighted least-squares (IRLS) updates, candidate screening, shortlisting, and final refinement are used to obtain a high-quality numerical solution without exhaustively searching all local minima.

The loss-function calibration used by the package combines a high-breakdown M-scale with approximately 95 percent Gaussian efficiency in the classical independent-error regression setting. This percentage is a tuning calibration and should not be interpreted as an exact efficiency statement for transformed panel data, where within-unit dependence may be present.

Value

If `effect = "fe"`, an object of classes `"fast_tau_fe_fit"` and `"fast_tau_panel_fit"` is returned. The object is the result of `fast_tau_fe`.

If `effect = "re"`, an object of classes `"fast_tau_re_fit"` and `"fast_tau_panel_fit"` is returned. The object is the result of `fast_tau_re`.

In both cases, the returned object contains at least the estimated regression coefficient vector β , together with fitting information specific to the selected panel specification.

Examples

```
## Generate a small fixed-effects panel
dat_fe <- generate_panel_data(
  N_ = 60,
  T_ = 4,
  K = 4,
  effect_model = "fe",
  contam_scheme = "none"
)

## Fit through the unified interface.
fit_fe <- fast_tau_panel(
  Y = dat_fe$Y,
  X = dat_fe$X,
  effect = "fe",
  N = 100,
  tt = 5
)

fit_fe$beta

## Generate a small random-effects panel
dat_re <- generate_panel_data(
  N_ = 60,
  T_ = 4,
  K = 4,
  effect_model = "re",
  contam_scheme = "none"
)

fit_re <- fast_tau_panel(
  Y = dat_re$Y,
  X = dat_re$X,
  effect = "re",
  N = 100,
  tt = 5
)

fit_re$beta

## Structural-mean predictions
pred_fe <- predict(fit_fe, newdata = dat_fe$X)
pred_re <- predict(fit_re, newdata = dat_re$X)
```

fast_tau_re

*Fast Tau Estimation for Random-Effects Panel Models***Description**

Fits the proposed robust fast Tau estimator for a balanced random-effects panel regression model. The procedure combines a first-difference fast-Tau pilot fit, robust scale-based variance-component estimation, residual-based observation weights, robust quasi-demeaning, and a final fast-Tau regression.

Usage

```
fast_tau_re(
  Y,
  X,
  N = 2000,
  kk = 2,
  tt = 20,
  rr = 2,
  approximate = 0,
  cluster_start = TRUE,
  m_cluster = NULL,
  seed = NULL
)
```

Arguments

Y	A numeric $n \times T$ matrix containing the panel responses, where n is the number of cross-sectional units and T is the number of time periods. The panel must be balanced.
X	A numeric $n \times T \times p$ array containing the regressors, where p is the number of regression coefficients. The first two dimensions must agree with those of Y.
N	Number of randomized starting subsamples used by each application of the fast-Tau algorithm. The default is 2000. This quantity corresponds to the number of random starts, denoted by S in the methodological description, and should not be confused with the number of cross-sectional units.
kk	Number of preliminary iteratively reweighted least-squares (IRLS) updates applied to each random or diagnostic starting candidate. The default is 2, corresponding to $\kappa = 2$.
tt	Number of the most promising candidate solutions retained for subsequent refinement. The default is 20, corresponding to the shortlist size $\ell = 20$.
rr	Number of fixed-point scale updates used when an approximate residual M-scale calculation is requested. The default is 2. This argument is relevant when <code>approximate</code> is nonzero.
approximate	Indicator controlling calculation of the residual M-scale during candidate evaluation. The default 0 uses the full scale calculation implemented by the fitting routine. A nonzero value uses the shorter approximation based on <code>rr</code> scale updates.

cluster_start	Logical value indicating whether cluster-diverse starting subsamples are used. The default is TRUE. When enabled, at most one transformed observation from any selected cross-sectional unit enters a starting subsample.
m_cluster	Number of distinct cross-sectional units used to form each cluster-diverse starting subsample. If NULL, the number of selected units is set equal to the number of regression coefficients, yielding a cluster-diverse elemental start. Larger values give overdetermined cluster-diverse starts, subject to the number of available units.
seed	Optional integer seed used for randomized starting-subsample generation. The default is NULL. The seed is handled locally within the fast-Tau computation.

Details

Consider the balanced random-effects panel regression model

$$y_{it} = x_{it}^T \beta + \alpha_i + \varepsilon_{it}, \quad i = 1, \dots, n, \quad t = 1, \dots, T,$$

where β is the common regression coefficient vector, α_i is an individual-specific random effect, and ε_{it} is the idiosyncratic error.

The random-effects specification requires the individual effects to satisfy the corresponding mean-independence condition with respect to the regressor history. Under the classical variance-components model,

$$\text{Var}(\alpha_i) = \sigma_\alpha^2, \quad \text{Var}(\varepsilon_{it}) = \sigma_\varepsilon^2,$$

the conventional quasi-demeaning parameter is

$$\theta = 1 - \left\{ \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + T\sigma_\alpha^2} \right\}^{1/2}.$$

The proposed estimator replaces the classical preliminary variance-component and unit-centering steps by robust counterparts.

The procedure consists of two fast-Tau regression stages connected by robust variance-component estimation and robust quasi-demeaning.

- **Pilot fast-Tau fit.** The individual effects are first eliminated by first differencing,

$$\Delta y_{it} = \Delta x_{it}^T \beta + \Delta \varepsilon_{it}.$$

The fast-Tau estimator is applied to this first-differenced sample, producing a robust pilot coefficient vector $\hat{\beta}_\tau^{(0)}$.

- **Robust scale-based variance components.** Pilot residuals are reconstructed on the original level scale,

$$r_{it}^{(0)} = y_{it} - x_{it}^T \hat{\beta}_\tau^{(0)}.$$

Pairwise residual differences and sums are formed over distinct time points. Let q_D and q_S denote normal-calibrated Q_n robust scales of the pooled residual differences and residual sums, respectively. The robust component estimates are

$$\hat{\sigma}_\varepsilon^2 = \max \left\{ \frac{q_D^2}{2}, \delta \right\},$$

and

$$\hat{\sigma}_\alpha^2 = \max \left\{ \frac{q_S^2 - q_D^2}{4}, \delta \right\},$$

where a small positive lower bound is used for numerical stability.

- **Robust quasi-demeaning.** The estimated quasi-demeaning parameter is

$$\hat{\theta} = 1 - \left\{ \frac{\hat{\sigma}_\varepsilon^2}{\hat{\sigma}_\varepsilon^2 + T\hat{\sigma}_\alpha^2} \right\}^{1/2},$$

restricted numerically to the interval $[0, 1]$.

To reduce the sensitivity of unit-specific centering to isolated contamination, the pilot residuals are centered within each unit and converted to bounded, redescending Tukey-bisquare weights. Using weights ϖ_{it} , weighted unit means are formed as

$$\bar{y}_{i,\varpi} = \frac{\sum_t \varpi_{it} y_{it}}{\sum_t \varpi_{it}}, \quad \bar{x}_{i,\varpi} = \frac{\sum_t \varpi_{it} x_{it}}{\sum_t \varpi_{it}}.$$

The robustly quasi-demeaned observations are then

$$y_{it}^\dagger = y_{it} - \hat{\theta} \bar{y}_{i,\varpi}, \quad x_{it}^\dagger = x_{it} - \hat{\theta} \bar{x}_{i,\varpi}.$$

- **Final fast-Tau fit.** The fast-Tau estimator is applied a second time to $(y_{it}^\dagger, x_{it}^\dagger)$. The resulting coefficient vector is returned as the final random-effects fast-Tau estimate.

The same multi-start fast-Tau settings are used in both the pilot and final regression stages. The Tau estimator minimizes a robust Tau-scale constructed from an M-scale and a second bounded loss function. This permits the robustness and efficiency components of the regression criterion to be calibrated separately.

Because the Tau objective is non-convex, the numerical algorithm uses multiple cluster-diverse starting values, a small number of preliminary IRLS refinements, screening and shortlisting, followed by more extensive refinement of promising candidates. The default configuration uses 2000 random starts, two preliminary IRLS updates, and a shortlist of 20 candidates.

The regressor matrix supplied to each fast-Tau stage is internally rescaled columnwise to improve numerical conditioning. Coefficients are transformed back to the original regressor scale before being returned.

Under the Gaussian random-effects model, the normal calibration of the Q_n scales makes the quantities $\hat{\sigma}_\varepsilon^2$ and $\hat{\sigma}_\alpha^2$ target the corresponding classical variance components. Under non-Gaussian distributions, squared robust scales need not equal second moments. In that setting, these quantities should be interpreted as robust, normal-calibrated scale-based analogues of the variance components rather than necessarily as consistent estimators of the population variances.

Value

An object of classes "fast_tau_re_fit" and "fast_tau_panel_fit". It is a list containing:

beta	Numeric vector containing the final fast-Tau estimates of the regression coefficients on the original regressor scale.
sigma2_a	Robust scale-based estimate of the individual-effect component σ_α^2 . Under the Gaussian RE model this targets the classical individual-effect variance component.
sigma2_e	Robust scale-based estimate of the idiosyncratic component σ_ε^2 . Under the Gaussian RE model this targets the classical idiosyncratic-error variance component.
theta	Estimated robust quasi-demeaning parameter,

$$\hat{\theta} = 1 - \left\{ \frac{\hat{\sigma}_\varepsilon^2}{\hat{\sigma}_\varepsilon^2 + T\hat{\sigma}_\alpha^2} \right\}^{1/2}.$$

fit	A list containing additional fitting information: • final: the final fast-Tau fit obtained from the robustly quasi-demeaned panel; • preliminary: the first-difference fast-Tau pilot fit; • beta_initial: the coefficient vector obtained from the pilot stage; • transformed_data: the robustly quasi-demeaned data supplied to the final fast-Tau fit. This object includes the transformed response and regressors, unit identifiers, observation-specific weights, weighted unit means, the robust residual scale, and the quasi-demeaning parameter.
effect	Character string equal to "random", identifying the fitted model as a random-effects specification.

Examples

```
## Generate a small random-effects panel
dat <- generate_panel_data(
  N_ = 60,
  T_ = 4,
  K = 4,
  effect_model = "re",
  contam_scheme = "none"
)

## Fit the proposed random-effects estimator.
fit <- fast_tau_re(
  Y = dat$Y,
  X = dat$X,
  N = 100,
  tt = 5
)

## Final regression coefficients
fit$beta

## Robust scale-based variance components
fit$sigma2_a
fit$sigma2_e

## Robust quasi-demeaning parameter
fit$theta

## Structural-mean predictions
pred <- predict(
  fit,
  newdata = dat$X
)

dim(pred)
```

generate_panel_data *Generate Balanced Panel Data with Optional Contamination*

Description

Generates balanced linear panel data under either a random-effects or correlated fixed-effects specification. The function supports Gaussian, Student-*t*, and chi-squared idiosyncratic errors together with several cellwise and blockwise vertical and leverage contamination mechanisms.

Usage

```
generate_panel_data(
  N_ = 100,
  T_ = 3,
  K = 4,
  distribution = "normal",
  beta = c(-3.4, 1.5, 2.8, -1.2),
  sigma2_a = 1,
  sigma2_e = 1,
  effect_model = c("re", "fe"),
  alpha_corr = 0.6,
  gamma_alpha = NULL,
  contam_scheme = c(
    "none",
    "random_vertical",
    "block_vertical",
    "random_leverage",
    "block_leverage",
    "random_vertical_leverage"
  ),
  eps_mean = 10,
  eps_std = 1,
  cont = 0.05,
  mu_leverage = 5,
  std_leverage = 1,
  block_fraction = 0.5,
  chi_df = NULL
)
```

Arguments

N_	Number of cross-sectional units. The default is 100.
T_	Number of time periods observed for each cross-sectional unit. The default is 3. The generated panel is balanced.
K	Number of regressors. The default is 4. The length of beta must equal K.
distribution	Character string specifying the idiosyncratic-error distribution. Available choices are "normal", "t", and "chi". For "normal", errors are generated from a centered Gaussian distribution with variance sigma2_e. For "t", errors are generated from a Student- <i>t</i> distribution with 5 degrees of freedom. For "chi", errors are generated from an uncentered chi-squared distribution with chi_df degrees of freedom.

beta	Numeric vector of regression coefficients. Its length must equal K. The default is $c(-3.4, 1.5, 2.8, -1.2)$.
sigma2_a	Variance parameter controlling the magnitude of the individual-specific effects. The default is 1.
sigma2_e	Idiosyncratic-error variance used when <code>distribution = "normal"</code> . The default is 1.
effect_model	Character string specifying the individual-effect design. Use "re" for random effects or "fe" for correlated fixed effects. The default is "re".
alpha_corr	Target correlation parameter used in the fixed-effects Mundlak-style construction. The default is 0.6. Internally, the value is restricted to the interval $[-0.99, 0.99]$. This argument is ignored when <code>effect_model = "re"</code> .
gamma_alpha	Optional numeric vector of length K defining the direction of the Mundlak index used to generate correlated fixed effects. The vector is normalized internally to have Euclidean norm one. If NULL and $K = 4$, the default direction is $c(1.0, -0.7, 0.5, 0.3)$. For other values of K, a decreasing sequence from 1 to 0.4 is used. This argument is relevant only when <code>effect_model = "fe"</code> .
contam_scheme	Character string specifying the contamination mechanism. Available choices are: "none", "random_vertical", "block_vertical", "random_leverage", "block_leverage", and "random_vertical_leverage". See Details for the precise construction.
eps_mean	Mean of the Gaussian distribution used to replace idiosyncratic errors under vertical contamination. The default is 10.
eps_std	Standard deviation of the Gaussian distribution used to replace idiosyncratic errors under vertical contamination. The default is 1.
cont	Target fraction of the $N \times T$ unit-time observations to contaminate. The default is 0.05. For random cellwise contamination, approximately $\text{round}(\text{cont} * N * T)$ observations are selected without replacement. Under block contamination, the achieved fraction may differ slightly because complete time blocks are contaminated.
mu_leverage	Mean of each regressor component used to generate leverage contamination. The default is 5.
std_leverage	Standard deviation of each regressor component used to generate leverage contamination. The default is 1.
block_fraction	Fraction of the time periods included in the consecutive contamination block for each selected unit under "block_vertical" and "block_leverage". The block length is $\max\{1, \lceil fT \rceil\}$ where f is specified by <code>block_fraction</code> . The default is 0.5.
chi_df	Degrees of freedom of the chi-squared idiosyncratic-error distribution when <code>distribution = "chi"</code> . If NULL, the function uses 2 for the fixed-effects design and 1 for the random-effects design. A user-supplied value must be a positive finite scalar.

Details

The generated data follow the balanced static panel model

$$y_{it} = x_{it}^T \beta + \alpha_i + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T,$$

where x_{it} is a K -dimensional regressor vector, β is the common slope vector, α_i is the individual-specific effect, and ε_{it} is the idiosyncratic error.

Regressor generation

The regressor array has dimensions $N \times T \times K$. Regressors are initially generated independently from standard normal distributions, except for the first regressor, which is replaced by

$$x_{it1} = w_{it} - 2, \quad w_{it} \sim \chi_2^2.$$

Thus, the first regressor is right-skewed but centered at zero, whereas the remaining regressors are standard Gaussian.

No intercept column is included in the generated design.

Random-effects design

When `effect_model = "re"`, the individual effects are generated independently of the regressors as

$$\alpha_i \sim N(0, \sigma_\alpha^2),$$

where `sigma2_a` specifies σ_α^2 . This corresponds to the standard random-effects requirement that the individual effect be independent of, and hence mean independent of, the regressor history in the data-generating mechanism.

Fixed-effects design

When `effect_model = "fe"`, the individual effects are explicitly correlated with the regressor history through a Mundlak-style construction.

Let

$$\bar{x}_i = \frac{1}{T} \sum_{t=1}^T x_{it}$$

denote the vector of unit-specific regressor means. After normalizing `gamma_alpha` to unit Euclidean norm, define the standardized Mundlak index

$$z_i = \text{standardize}(\gamma_\alpha^\top \bar{x}_i).$$

An independent Gaussian vector is generated, orthogonalized with respect to z , and standardized to produce η_i . The fixed effect is then constructed as

$$\alpha_i = \sigma_\alpha \left\{ \rho z_i + \sqrt{1 - \rho^2} \eta_i \right\},$$

where ρ is the value of `alpha_corr` restricted to $[-0.99, 0.99]$.

The orthogonalization makes the realized dependence between the individual effects and the Mundlak index closely controlled by `alpha_corr`, while `sigma2_a` determines the scale of the individual effects. This construction is used to generate an FE setting in which the individual effects are intentionally correlated with the regressors.

Idiosyncratic errors

Before contamination, the idiosyncratic errors are generated according to distribution:

- "normal":

$$\varepsilon_{it} \sim N(0, \sigma_\varepsilon^2),$$

where σ_ε^2 is specified by `sigma2_e`.

- "t":

$$\varepsilon_{it} \sim t_5.$$

The current implementation does not rescale the Student- t draws using `sigma2_e`.

- "chi":

$$\varepsilon_{it} \sim \chi^2_{\nu}.$$

The degrees of freedom ν are specified by `chi_df`. The chi-squared errors are deliberately uncentered. The current implementation does not rescale these draws using `sigma2_e`.

Contamination schemes

Contamination is applied after the clean regressors and individual effects have been generated.

- "none": no observations are contaminated.
- "random_vertical": approximately a fraction `cont` of unit-time cells is sampled uniformly without replacement. At each selected cell, the original idiosyncratic error is replaced by an independent draw from a Gaussian distribution with mean `eps_mean` and standard deviation `eps_std`.
- "block_vertical": selected cross-sectional units receive vertical contamination over a randomly positioned consecutive block of time periods. The block length is determined by `block_fraction`. Errors in those cells are replaced by draws from a Gaussian distribution with mean `eps_mean` and standard deviation `eps_std`.
- "random_leverage": approximately a fraction `cont` of unit-time cells is sampled uniformly without replacement. At every selected cell, all K observed regressor components are replaced independently by draws from independent Gaussian draws with mean `mu_leverage` and standard deviation `std_leverage`.
- "block_leverage": leverage contamination is introduced over randomly positioned consecutive blocks within selected cross-sectional units. All regressor components in a contaminated cell are replaced by draws from independent Gaussian draws with mean `mu_leverage` and standard deviation `std_leverage`.
- "random_vertical_leverage": the same randomly selected unit-time cells receive both forms of contamination. Their idiosyncratic errors are replaced by draws from $N(\text{eps_mean}, \text{eps_std}^2)$, and their observed regressor vectors are replaced by Gaussian leverage draws.

For all schemes involving leverage contamination, the response is generated from the *uncontaminated* regressor vector rather than from the contaminated value subsequently observed by the estimator. Consequently, the leverage and combined contamination schemes create bad leverage observations: the contaminated carrier does not remain on the regression surface.

The structural component returned as `MU` is therefore

$$\mu_{it} = x_{it,\text{clean}}^T \beta$$

for leverage-contaminated designs. Under pure vertical contamination or no contamination, the observed and response-generating regressors coincide.

Finally, responses are generated as

$$y_{it} = \mu_{it} + \alpha_i + \varepsilon_{it}.$$

Value

A list containing:

X	Numeric array of dimensions $N \times T \times K$ containing the regressors observed by the estimator. Under leverage contamination, selected entries differ from the clean regressors used to generate the corresponding responses.
Y	Numeric $N \times T$ matrix containing the generated panel responses.

MU	Numeric $N \times T$ matrix containing the structural regression component $x_{it}^T \beta$ used to generate the response. For leverage-contaminated cells, this is calculated from the uncontaminated regressor vector.
alpha	Numeric $N \times 1$ matrix containing the generated individual-specific effects.
epsilon	Numeric $N \times T$ matrix containing the final idiosyncratic errors after any vertical contamination has been applied.
outlier_cells	Integer vector identifying contaminated unit-time cells. Indices are stored in unit-major order, so that cell (i, t) corresponds to $(i - 1) * T_+ + t$. An empty integer vector is returned when no contamination is applied.
outlier_units	Sorted integer vector identifying cross-sectional units containing at least one contaminated cell. An empty integer vector is returned when no contamination is applied.
gamma_alpha	Normalized coefficient vector defining the Mundlak index in the fixed-effects design. For the random-effects design this is typically NULL.
effect_model	Character string identifying the generated individual-effect specification: "fe" or "re".
chi_df	Degrees of freedom used for chi-squared idiosyncratic errors. The value is returned even when a different error distribution is used.

Examples

```
## Clean random-effects panel
dat_re <- generate_panel_data(
  N_ = 50,
  T_ = 4,
  K = 4,
  effect_model = "re",
  contam_scheme = "none"
)

dim(dat_re$X)
dim(dat_re$Y)
dat_re$effect_model

## Correlated fixed-effects panel
dat_fe <- generate_panel_data(
  N_ = 50,
  T_ = 4,
  K = 4,
  effect_model = "fe",
  alpha_corr = 0.6,
  contam_scheme = "none"
)

dat_fe$gamma_alpha

## Random vertical contamination
dat_vertical <- generate_panel_data(
  N_ = 50,
  T_ = 5,
```

```

    K = 4,
    effect_model = "fe",
    contam_scheme = "random_vertical",
    cont = 0.10
  )

  length(dat_vertical$outlier_cells)
  dat_vertical$outlier_units

## Bad-leverage contamination
dat_leverage <- generate_panel_data(
  N_ = 50,
  T_ = 5,
  K = 4,
  effect_model = "re",
  contam_scheme = "random_leverage",
  cont = 0.10
)

length(dat_leverage$outlier_cells)

```

predict.fast_tau_fe_fit

Predict from a Fast Tau Fixed-Effects Model

Description

Computes structural-mean predictions from a fitted fast Tau fixed-effects panel model. Predictions are obtained from the estimated common regression component, $x^T \hat{\beta}_\tau$, and do not include an estimated individual fixed effect.

Usage

```
## S3 method for class fast_tau_fe_fit
predict(object, newdata, ...)
```

Arguments

object	A fitted object of class "fast_tau_fe_fit", typically returned by fast_tau_fe or by fast_tau_panel with effect = "fe".
newdata	New predictor values on the original regressor scale. The argument may be supplied in one of three forms: • a numeric vector of length p, representing one predictor vector; • a numeric matrix or data frame with p columns, representing multiple predictor vectors; • a numeric $N \times T \times p$ array, representing panel-structured predictor values for N units and T time periods. In all cases, p must equal the number of coefficients in the fitted model.
...	Additional arguments for compatibility with the generic predict interface. They are currently ignored.

Details

For the fixed-effects panel model

$$y_{it} = x_{it}^T \beta + \alpha_i + \varepsilon_{it},$$

the fitted fast Tau estimator provides an estimate $\hat{\beta}_\tau$ of the common regression coefficient vector. For a supplied predictor vector x , this method returns the structural component

$$\hat{\mu}(x) = x^T \hat{\beta}_\tau.$$

The prediction is evaluated using the regressors on their *original scale*. Users should therefore supply the original predictor values in `newdata`; first differences, within transformations, or other fixed-effects transformations should not be applied before calling `predict()`.

This distinction is important because the transformation used during estimation serves to eliminate the individual-specific effect and estimate the common slope vector. Once $\hat{\beta}_\tau$ has been estimated, structural prediction for new predictor values is naturally made on the original scale.

For an individual already observed in the fitting sample, a conditional fitted value can in principle be written as

$$\hat{y}_{it} = x_{it}^T \hat{\beta}_\tau + \hat{\alpha}_i,$$

where, for the proposed fixed-effects procedure, the accompanying methodology estimates the individual effect using the unitwise residual median,

$$\hat{\alpha}_i = \text{median}_t \left\{ y_{it} - x_{it}^T \hat{\beta}_\tau \right\}.$$

Such a quantity requires observed response history for that individual.

The present method is intentionally different: it uses only `newdata` and the estimated coefficient vector stored in `object`. Consequently, it returns $x^T \hat{\beta}_\tau$ only and does not estimate or add an individual-specific fixed effect. This makes the method appropriate for structural prediction and for predictor sets belonging to previously unobserved units.

The shape of the returned object follows the shape of `newdata`. No additional panel transformation is performed during prediction.

Value

Predicted structural means based on $x^T \hat{\beta}_\tau$. The returned object depends on the form of `newdata`:

- If `newdata` is a numeric vector of length p , a single numeric prediction is returned.
- If `newdata` is a numeric matrix or data frame with m rows and p columns, a numeric vector of length m is returned.
- If `newdata` is an $N \times T \times p$ numeric array, an $N \times T$ numeric matrix of predictions is returned.

For array input, available dimension names for the first two dimensions are preserved in the prediction matrix.

Examples

```
## Generate a fixed-effects panel

dat <- generate_panel_data(
  N_ = 60,
  T_ = 4,
```

```

    K = 4,
    effect_model = "fe",
    contam_scheme = "none"
  )

  ## Fit the proposed fixed-effects estimator.
  fit <- fast_tau_fe(
    Y = dat$Y,
    X = dat$X,
    N = 100,
    tt = 5
  )

  ## Panel-structured predictions
  pred_panel <- predict(
    fit,
    newdata = dat$X
  )

  dim(pred_panel)

  ## Prediction for one predictor vector
  x_new <- c(0.5, -0.2, 1.1, 0.7)

  predict(
    fit,
    newdata = x_new
  )

  ## Predictions for several predictor vectors
  X_new <- matrix(
    rnorm(20),
    nrow = 5,
    ncol = 4
  )

  predict(
    fit,
    newdata = X_new
  )

```

predict.fast_tau_re_fit

Predict from a Fast Tau Random-Effects Model

Description

Computes marginal structural-mean predictions from a fitted fast Tau random-effects panel model. Predictions are based on the estimated common regression component, $x^T \widehat{\beta}_{\tau, \text{RE}}$, and do not include a predicted individual-specific random effect.

Usage

```
## S3 method for class fast_tau_re_fit
predict(object, newdata, ...)
```

Arguments

- | | |
|---------|--|
| object | A fitted object of class "fast_tau_re_fit", typically returned by <code>fast_tau_re</code> or by <code>fast_tau_panel</code> with <code>effect = "re"</code> . |
| newdata | <p>New predictor values on the original regressor scale.</p> <p>The argument may be supplied in one of three forms:</p> <ul style="list-style-type: none"> • a numeric vector of length p, representing one predictor vector; • a numeric matrix or data frame with p columns, representing multiple predictor vectors; • a numeric $N \times T \times p$ array, representing panel-structured predictor values for N units and T time periods. <p>In every case, p must equal the number of regression coefficients in the fitted model.</p> |
| ... | Additional arguments included for compatibility with the generic <code>predict</code> interface. They are currently ignored. |

Details

Consider the random-effects panel regression model

$$y_{it} = x_{it}^T \beta + \alpha_i + \varepsilon_{it},$$

where α_i is the individual-specific random effect and ε_{it} is the idiosyncratic error.

The proposed random-effects estimator yields an estimate $\hat{\beta}_{\tau, \text{RE}}$ of the common regression coefficient vector. For a supplied predictor vector x , this method returns

$$\hat{\mu}(x) = x^T \hat{\beta}_{\tau, \text{RE}}.$$

Thus, the method predicts the structural or marginal regression component rather than an individual-specific conditional response.

The random-effects specification assumes that the individual effect has zero mean conditional on the relevant regressor information. For a previously unobserved unit with no response history, the realized random effect cannot be estimated from the new regressors alone. Its corresponding prediction is therefore based on

$$E(y_{\text{new}} \mid x_{\text{new}}) = x_{\text{new}}^T \beta,$$

and the fitted counterpart is

$$\hat{y}_{\text{new}} = x_{\text{new}}^T \hat{\beta}_{\tau, \text{RE}}.$$

For a unit already observed in the fitting sample, an individual-specific conditional prediction can additionally use a prediction of its random effect. In the accompanying methodology, the random effect is predicted using the plug-in shrinkage form

$$\hat{\alpha}_i = \hat{\lambda} \bar{r}_i,$$

where

$$\bar{r}_i = \frac{1}{T} \sum_{t=1}^T \left(y_{it} - x_{it}^T \hat{\beta}_{\tau, \text{RE}} \right)$$

and

$$\hat{\lambda} = \frac{\hat{\sigma}_\alpha^2}{\hat{\sigma}_\alpha^2 + \hat{\sigma}_\varepsilon^2 / T}.$$

Equivalently,

$$\hat{\lambda} = 1 - (1 - \hat{\theta})^2,$$

where $\hat{\theta}$ is the robust quasi-demeaning parameter returned by `fast_tau_re`.

A corresponding conditional fitted value for an observed unit is therefore

$$\hat{y}_{it} = x_{it}^T \hat{\beta}_{\tau, \text{RE}} + \hat{\alpha}_i.$$

The present `predict()` method does not perform this individual-effect prediction, because it receives only `newdata` and the fitted model object and does not use the response history required to construct $\bar{r}_i$. It therefore returns the structural component only.

The predictor values supplied through `newdata` must be on the original regressor scale. Users should not apply the robust quasi-demeaning transformation before prediction. That transformation is an estimation device used internally to obtain $\hat{\beta}_{\tau, \text{RE}}$; predictions from new regressors are evaluated on the original scale.

The shape of the returned result follows the shape of `newdata`.

Value

Predicted structural means based on $x^T \hat{\beta}_{\tau, \text{RE}}$. The form of the returned object depends on `newdata`:

- If `newdata` is a numeric vector of length p , a single numeric prediction is returned.
- If `newdata` is a numeric matrix or data frame with m rows and p columns, a numeric vector of length m is returned.
- If `newdata` is an $N \times T \times p$ numeric array, an $N \times T$ numeric matrix of predictions is returned.

For array input, available dimension names for the first two dimensions are preserved in the returned prediction matrix.

Examples

```
## Generate a random-effects panel

dat <- generate_panel_data(
  N_ = 60,
  T_ = 4,
  K = 4,
  effect_model = "re",
  contam_scheme = "none"
)

## Fit the proposed random-effects estimator.
fit <- fast_tau_re(
  Y = dat$Y,
  X = dat$X,
```

```
N = 100,
tt = 5
)

## Panel-structured structural predictions
pred_panel <- predict(
  fit,
  newdata = dat$X
)

dim(pred_panel)

## Prediction for one new predictor vector
x_new <- c(0.5, -0.2, 1.1, 0.7)

predict(
  fit,
  newdata = x_new
)

## Predictions for several new predictor vectors
X_new <- matrix(
  rnorm(20),
  nrow = 5,
  ncol = 4
)

predict(
  fit,
  newdata = X_new
)
```

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